Progress Report: Low Emotional Engagement Detection using Multimodal Sensor Data

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1 Introduction

Emotional engagement reflects an individual's cognitive and affective involvement during an experience, such as watching a video or interacting with a system. Accurately detecting low engagement is crucial for adaptive learning, mental health monitoring, and human-computer interaction.

In affective computing, emotional states are often represented using the **valence-arousal model**, where *valence* indicates the positivity or negativity of an emotion, and *arousal* reflects its intensity. In this study, user-provided valence and arousal scores were transformed into a binary label: `low engagement` (label 1) when both valence and arousal were below the median threshold, and `high engagement` (label 0) otherwise.

Identifying low engagement from multimodal sensor data—such as physiological signals and eye-tracking features—can enable the development of responsive systems that adjust content or interventions in real-time, enhancing user experience and outcomes.

2 Project Overview

This project aims to develop a multimodal system to detect low emotional engagement using physiological signals and eye-tracking data. The overall workflow spans from data collection and preprocessing to model development and deployment. As of now, foundational research, platform setup, pilot studies, and multi-user data collection have been completed. The upcoming phases will focus on training baseline models, expanding the dataset, and refining our detection approach for real-world deployment.

2.1 Exploratory Data Analysis

A preliminary analysis of the final dataset (1996 samples, 39 features) revealed a moderate class imbalance in the target variable `probe`, which encodes low emotional engagement. Approximately 67% of the samples are labeled as 0 (normal engagement), while the remaining 33% are labeled as 1 (low engagement). This imbalance should be considered during model training to avoid bias toward the majority class.

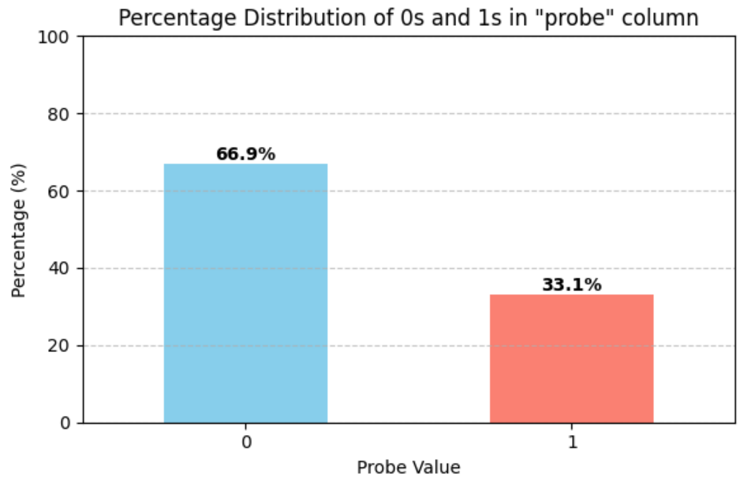

Figure 1: Percentage distribution of classes in the `probe` variable.

3 Methodology

3.1 Data Collection

Participants were asked to watch a set of 8 video clips selected to cover all four quadrants of the valence-arousal space. To minimize emotional carryover, each video was preceded by a 2-minute blue screen. Throughout the session, users continuously self-annotated their emotional state using an annotation app that allowed them to rate valence and arousal on a 1–9 scale.

Physiological data (GSR and HR) were recorded using a wrist-mounted sensor device, while eye-tracking and pupil diameter data were collected using the Pupil Labs Core system. All modalities were logged with timestamps and later aligned by aggregating data into fixed-length 5-second windows using time-range mapping.

3.2 Annotation and Labeling

Throughout the experiment, participants provided continuous self-reported annotations of their emotional states using a custom-built interface. These annotations captured **valence** (pleasantness) and **arousal** (intensity) on a scale of 1 to 9.

To generate a binary indicator of emotional engagement, we computed the median valence and arousal rating for each video per user. Any 5-second window where both valence and arousal scores fell below the corresponding user-specific medians was labeled as **Low Emotional Engagement** (label = 1). All other cases were considered as non-low engagement (label = 0).

This relative thresholding approach accounts for individual differences in emotional baselines and avoids the pitfalls of using static cutoff values across users.

3.3 Data Preprocessing

The dataset comprises multimodal streams collected during user interaction, including physiological signals (Galvanic Skin Response [GSR], Heart Rate [HR]), eye-tracking signals (pupil diameter, gaze vectors, blink rate), and self-reported engagement annotations. A multi-step preprocessing pipeline was implemented to clean, normalize, and synchronize these heterogeneous sources into a unified, analyzable format.

Signal Cleaning and Filtering: Each modality was independently cleaned. Physiological signals were denoised by removing NaNs, extreme outliers, and invalid entries. Eye-tracking data (including gaze vectors and pupil diameter) was filtered to retain only valid frames where the confidence scores exceeded a threshold (typically 0.6). Blinks and rapid head movements were also detected and encoded during this phase. Annotation data was filtered to retain valid numeric engagement labels.

Normalization: All continuous features across modalities were normalized using min-max scaling, applied individually for each user. This step ensured that variations in sensor calibration or individual baselines did not dominate learning, enabling fair cross-user comparisons. For example, pupil diameter from both 2D and 3D methods was scaled to the [0, 1] range per user session.

Windowing and Resampling: To enable consistent feature extraction across modalities with varying sampling frequencies, the data was segmented into fixed 5-second windows with sliding window aggregation. Within each window, statistical aggregates were computed (e.g., mean, standard deviation), capturing local temporal dynamics without overwhelming granularity. Only windows containing complete data across all modalities were retained.

Cross-Modal Synchronization: Temporal alignment between physiological signals and pupil/gaze data was achieved using system and synchronization timestamps extracted from the info.player.json files. Pupil timestamps in seconds were converted to system time in milliseconds, and then overlap-based merging was performed. This approach ensured that multimodal data points falling within the same 5-second temporal window were matched and merged. Engagement annotations were then assigned to the corresponding windows by matching time intervals.

This preprocessing pipeline produced a clean, synchronized, and windowed dataset, ready for multimodal feature extraction and modeling. Each row in the final dataset represents a 5-second time window containing aggregated physiological and eye-tracking features, alongside a timestamp and user-specific engagement label.

3.4 Feature Engineering

After preprocessing, features were extracted independently from each modality within each 5-second window. These features aimed to capture physiological arousal, attentional patterns, and user state indicators.

- **Physiological Features (GSR and HR):** For each window, the mean, standard deviation, and percent change were computed for both GSR and HR signals. These features reflect fluctuations in autonomic nervous system activity, capturing stress and arousal states relevant to engagement.

- **Pupil Diameter:** Features were computed from both the **pye3d 3D model** and the **2D c++** pupil detection method. For each, the **mean** and **standard deviation** of pupil size were calculated per eye. This redundancy improves robustness to noise and helps capture light response and cognitive load fluctuations.
- **Blink and Confidence Metrics:** From the eye-tracking stream, the **blink rate** (blinks per second) and **blink duration** were calculated per window. In addition, average and variance of **gaze confidence values** were extracted to reflect data quality and tracking stability.
- **Gaze and Head Movement:** Gaze coordinates (x, y) and normalized 3D vectors were used to compute **mean**, **standard deviation**, and **total displacement**, indicating attentional shifts and visual scanning behavior. Similarly, **head position** and **gaze origin** vectors were processed to quantify head movement using Euclidean distance between consecutive points.
- **Temporal Metadata:** Each window was tagged with a timestamp (start and end in milliseconds) and a unique user ID. Engagement labels were appended based on temporal overlap with user annotations.

3.5 Modeling and Interpretability

Tree-based classifiers — Random Forest, XGBoost, and LightGBM — were trained using the full multimodal feature set. For model interpretability, we applied SHAP (SHapley Additive exPlanations) to identify the most influential features. We also used PCA and correlation analysis to examine feature redundancy and clustering patterns across users.

3.6 Data Collection Setup

Participants interacted with a custom-built web platform designed for synchronized video playback and continuous emotion annotation. The system integrated multiple sensors and input streams, including:

- **Physiological Signals:** GSR and HR data were recorded using a wrist-mounted wearable device.
- **Eye-tracking:** Pupil diameter and gaze data were collected using the Pupil Labs Core eye-tracker with head-mounted cameras.
- **Annotation Interface:** Participants continuously rated their emotional state using an on-screen annotation app, providing valence and arousal scores on a 1–9 scale.

The stimuli consisted of 8 carefully selected video clips, two from each quadrant of the valence-arousal space (high/low valence $\times$ high/low arousal). Each video was preceded by a 2-minute blue screen to minimize emotional carryover effects. Full details of the stimulus set and selection criteria are available in the Appendix.

Data was collected from four participants (2 male, 2 female), all of whom provided informed consent before participating in the study. Users were instructed to watch all videos in a single session while wearing the sensors and providing continuous feedback via the annotation tool.

4 Feature Engineering and Analysis

4.1 Correlation Matrix

To understand the linear relationships between features and their relevance to the target variable, we computed the Pearson correlation matrix on the processed dataset. This allows us to identify which features are most linearly associated with the target label, `probe`, representing low emotional engagement.

Figure 2 shows the full feature-feature correlation heatmap. It highlights clusters of interrelated features, particularly among eye-tracking variables such as 3D pupil diameters, ellipse dimensions, and gaze metrics.

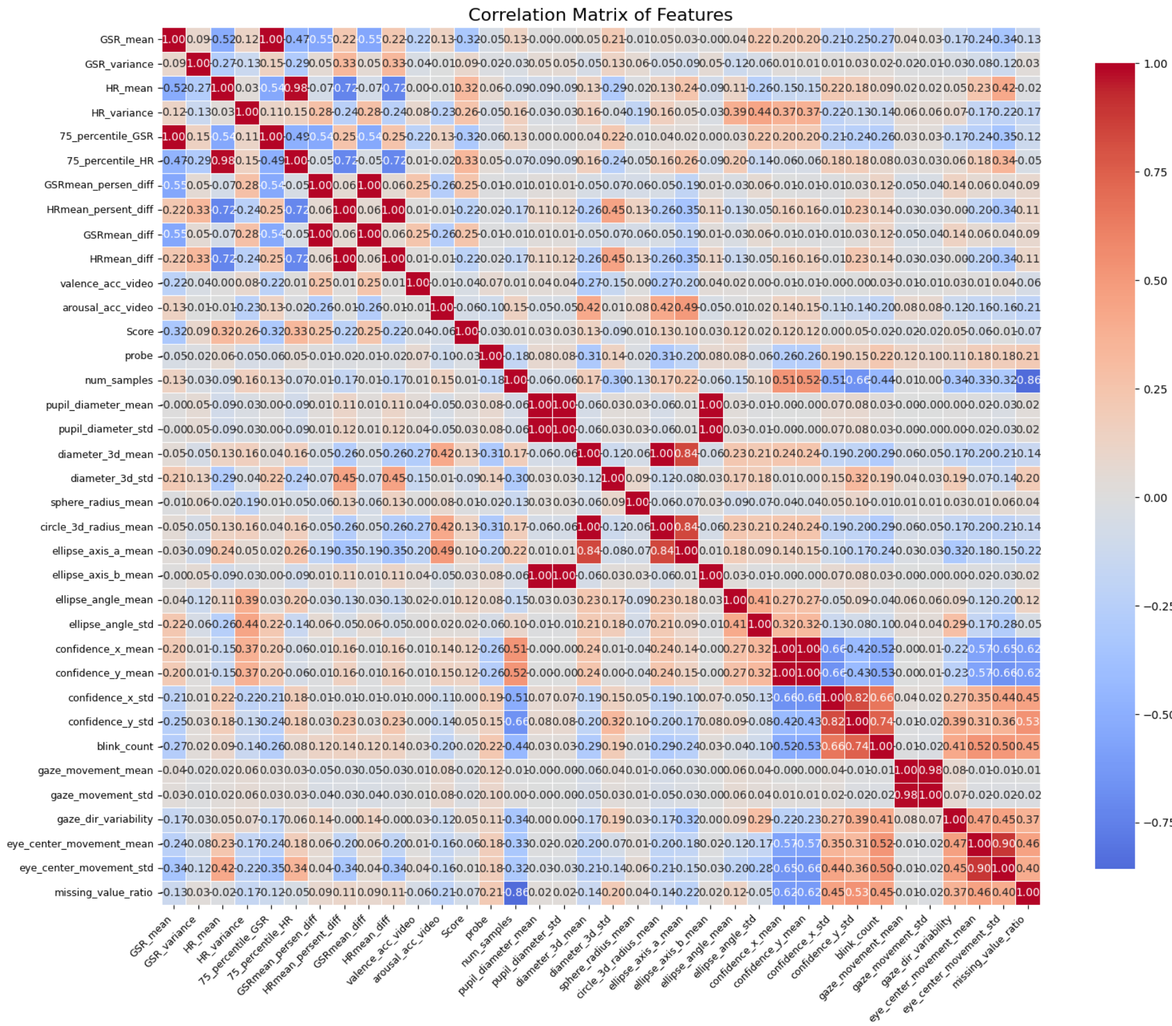

Figure 2: Pearson Correlation Matrix across all features

We further analyzed the individual correlation of each feature with the target variable. The top 10 most correlated features with `probe` (by absolute value) are shown in Figure 3.

Notably, features like `circle_3d_radius_mean` and `diameter_3d_mean` showed the highest negative correlation (0.31) with low emotional engagement, while `blink_count` and `missing_value_ratio` had the strongest positive correlations (+0.21).

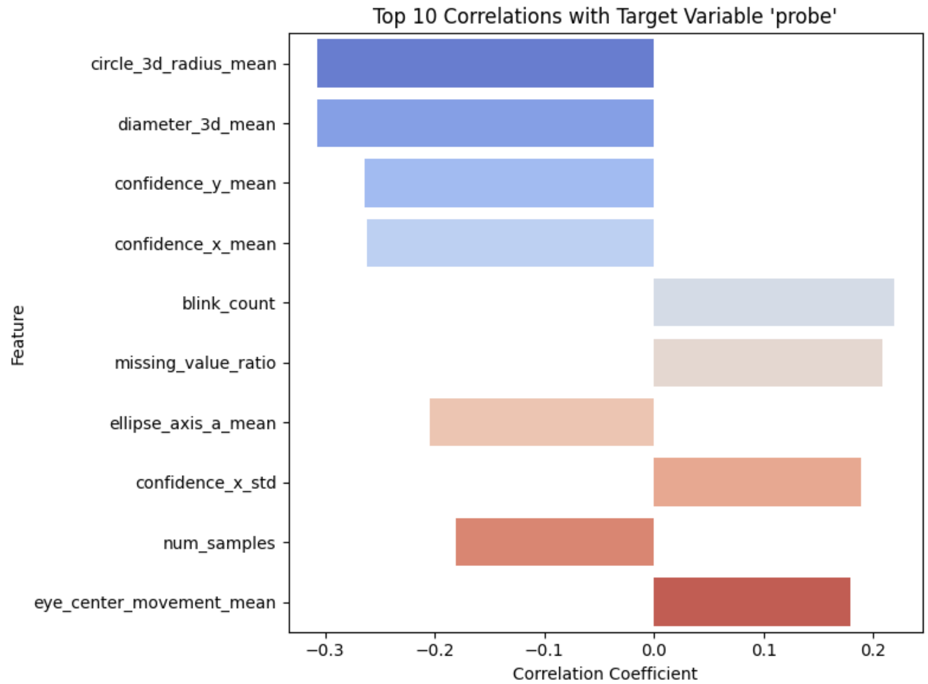

Figure 3: Top 10 Features Most Correlated with the Target Label (`probe`)

This analysis suggests that several eye-tracking metrics (particularly 3D diameter and confidence measures) play a more significant role in identifying low emotional engagement compared to physiological signals like GSR and HR, which exhibited weaker correlations with the target.

4.2 Tree-Based Feature Importance

To understand which features contribute most to predicting low emotional engagement, we employed three popular tree-based classifiers: **Random Forest**, **XGBoost**, and **LightGBM**. These models inherently provide feature importance scores based on different heuristics such as split gain, information gain, and frequency.

Random Forest

We trained a Random Forest classifier on the preprocessed data. The top 20 features ranked by importance are shown in Figure 4. Notably, `diameter_3d_mean` and `circle_3d_radius_mean` were among the top contributors.

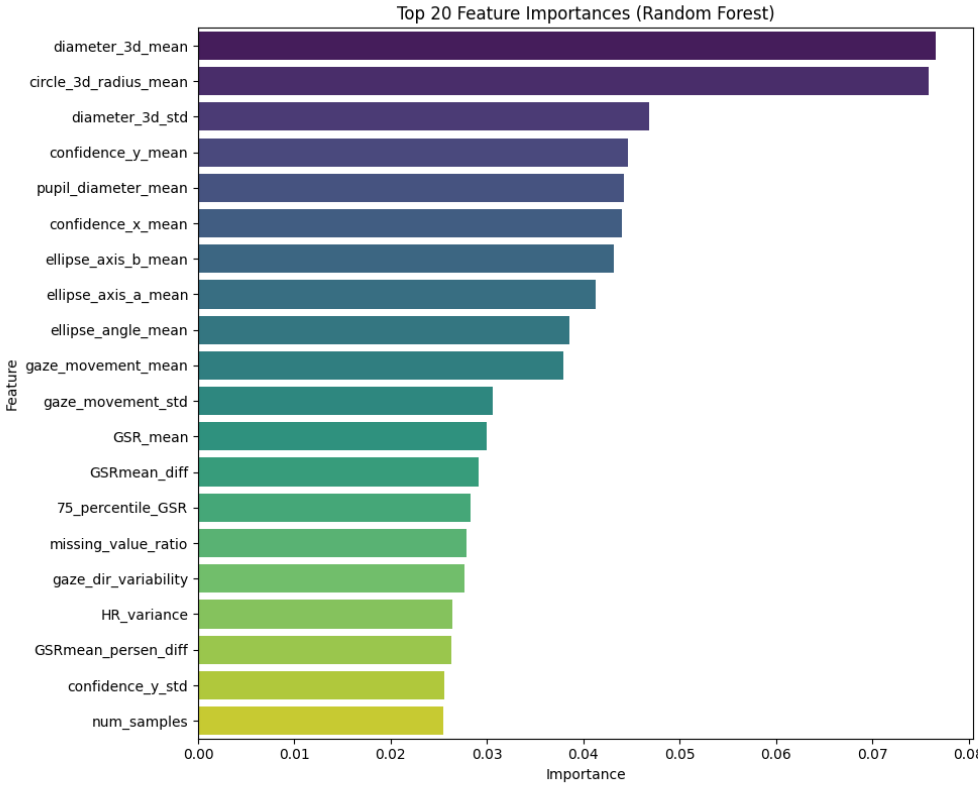

Figure 4: Top 20 Feature Importances using Random Forest

The classifier achieved an accuracy of **95%** on the test set. Class distribution in training data was approximately 67% for class 0 and 33% for class 1, indicating a mild class imbalance.

XGBoost and LightGBM

We further trained two gradient boosting classifiers — XGBoost and LightGBM — to cross-validate the importance rankings.

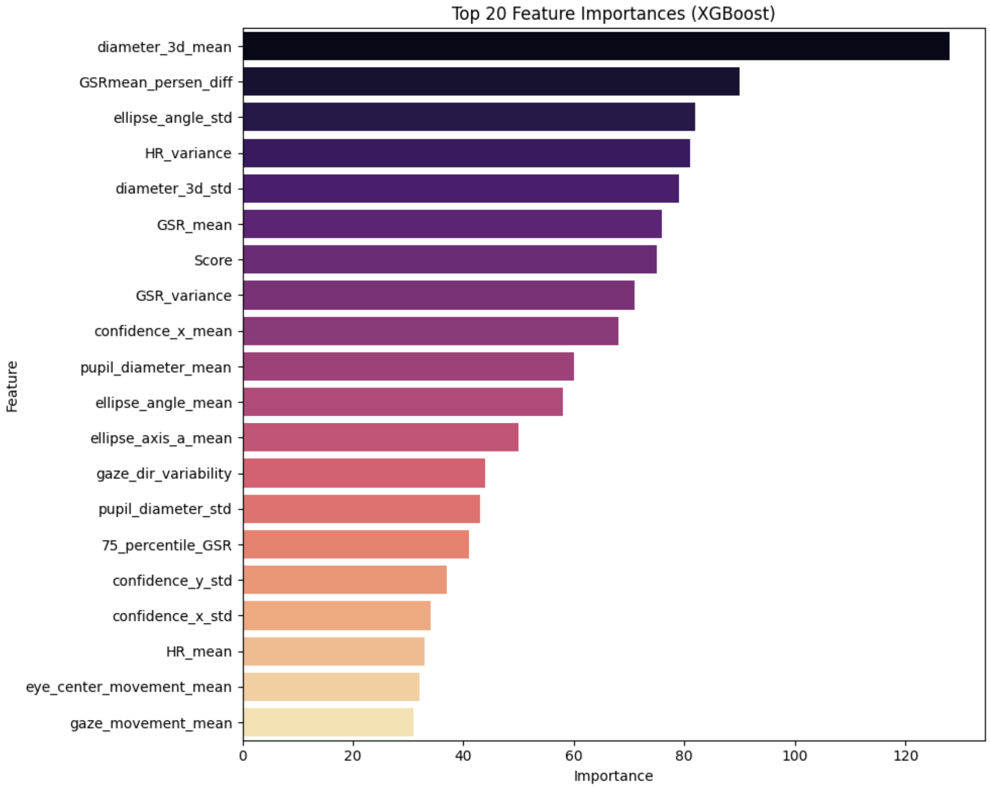

Figure 5: Top 20 Feature Importances using XGBoost

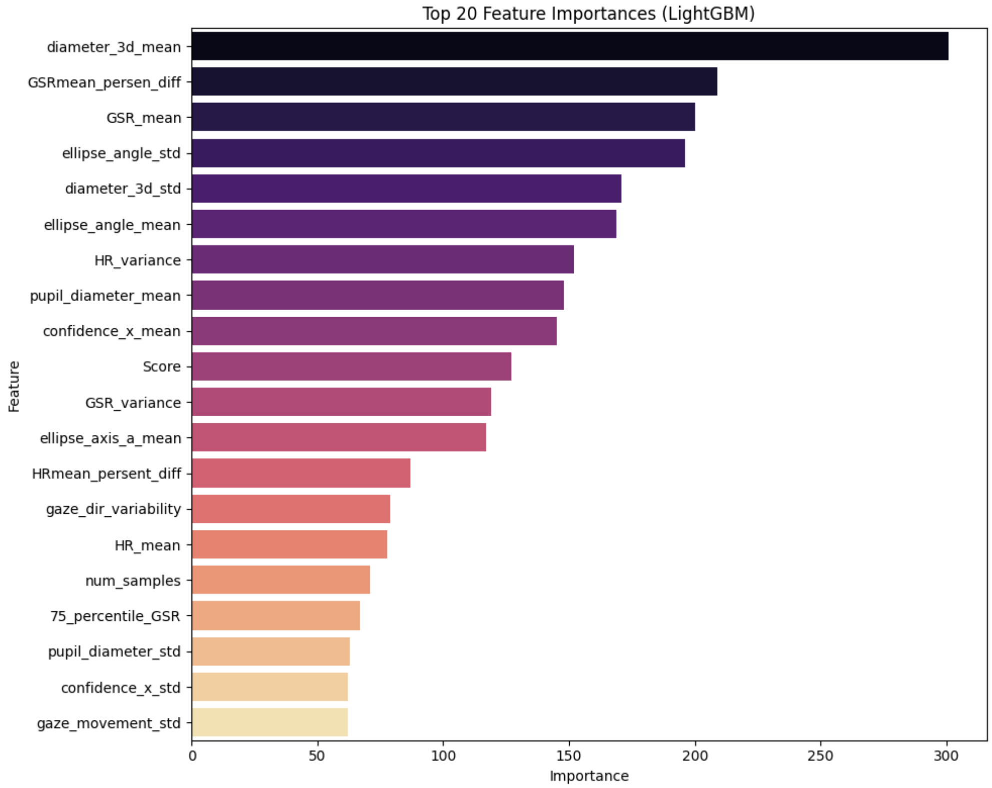

Figure 6: Top 20 Feature Importances using LightGBM

Table 1 summarizes the top 10 features across all three models:

Table 1: Top Features by Model

XGBoost	**LightGBM**	**Random Forest**
diameter_3d_mean	diameter_3d_mean	diameter_3d_mean
GSRmean_persen_diff	GSRmean_persen_diff	circle_3d_radius_mean
ellipse_angle_std	GSR_mean	diameter_3d_std
HR_variance	ellipse_angle_std	confidence_y_mean
diameter_3d_std	ellipse_angle_mean	pupil_diameter_mean
GSR_mean	HR_variance	confidence_x_mean
Score	pupil_diameter_mean	ellipse_axis_b_mean
GSR_variance	confidence_x_mean	ellipse_axis_a_mean
confidence_x_mean	Score	ellipse_angle_mean
pupil_diameter_mean	Score	gaze_movement_mean

Both XGBoost and LightGBM achieved an accuracy of **96%**, slightly outperforming Random Forest. There is consistent agreement across all models on the importance of features related to 3D pupil diameter, GSR, and ellipse geometry, suggesting these are strong indicators of low emotional engagement.

4.3 Tree-Based Feature Importance via SHAP

To quantify and interpret feature contributions in predictive modeling, we applied SHAP (SHapley Additive exPlanations) to tree-based classifiers: LightGBM, XGBoost, and Random Forest. SHAP provides both global and local interpretability by assigning importance scores to features based on their marginal contribution to the model output.

LightGBM Results

Figure 7 presents the top features ranked by their mean absolute SHAP values for the LightGBM model. Notably, `diameter_3d_mean` emerged as the most impactful feature, followed by `confidence_x_mean`, `diameter_3d_std`, and `GSR_mean`.

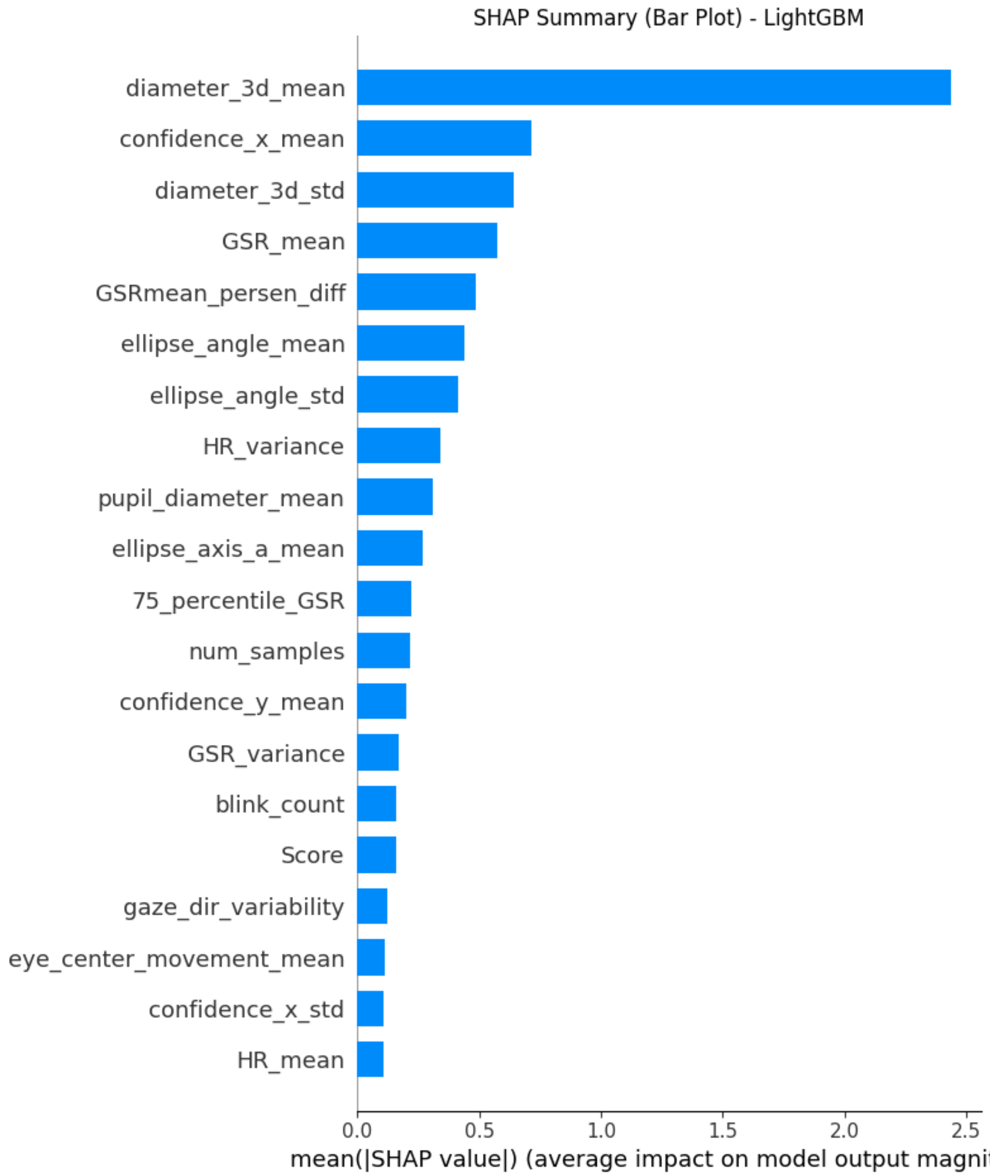

Figure 7: SHAP Summary (Bar Plot) for LightGBM

To further explore how individual feature values affect predictions, we plotted the SHAP beeswarm plot in Figure 8. This highlights the distribution and influence of feature values across the dataset.

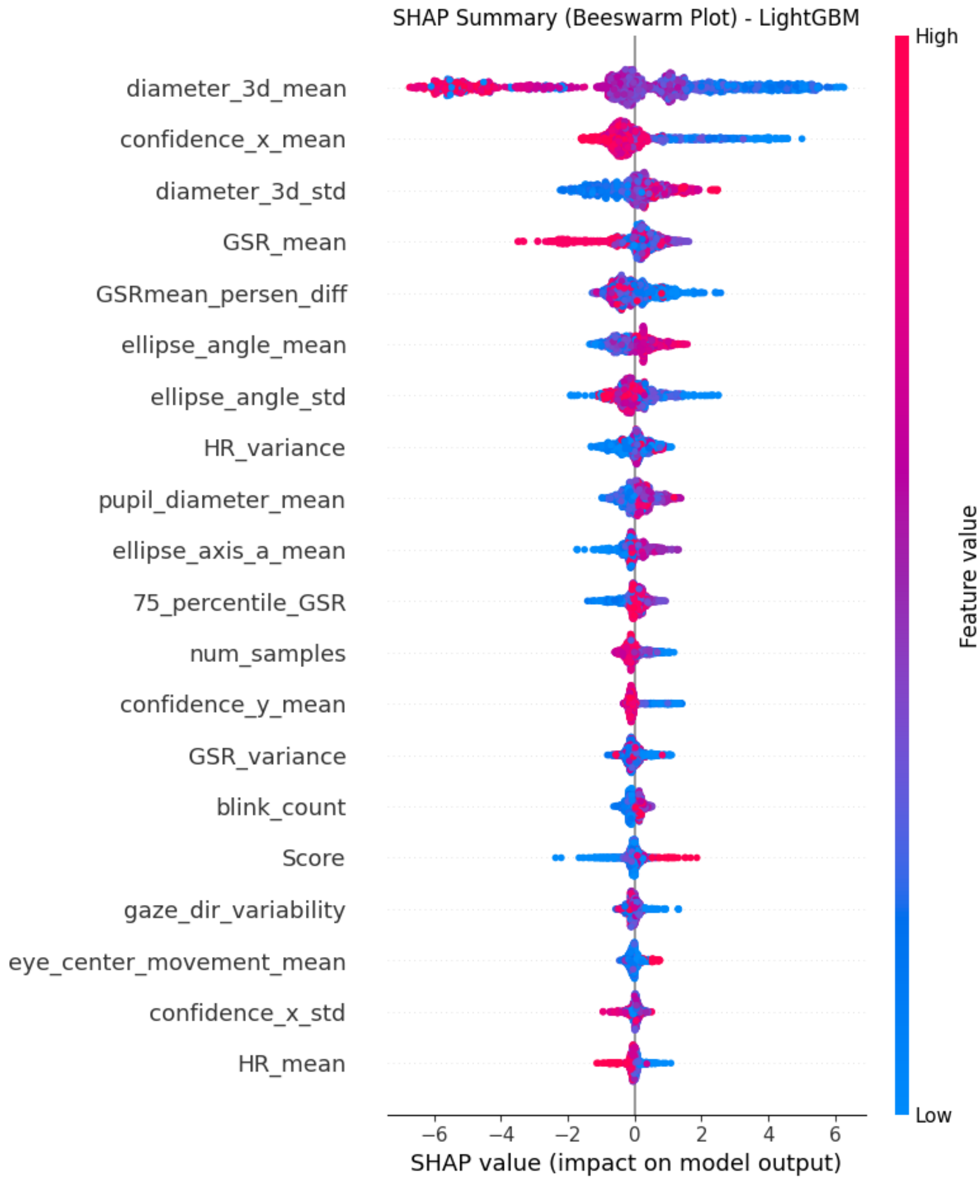

Figure 8: SHAP Beeswarm Plot for LightGBM

Top 5 SHAP-ranked Features (LightGBM):

- `diameter_3d_mean`
- `confidence_x_mean`
- `diameter_3d_std`
- `GSR_mean`
- `GSRmean_persen_diff`

XGBoost Results

Figures 9 and 10 show similar plots for the XGBoost model. The top contributing features remain consistent, particularly `diameter_3d_mean`, `GSR_mean`, and `confidence_x_mean`, reaffirming their significance in predicting low engagement.

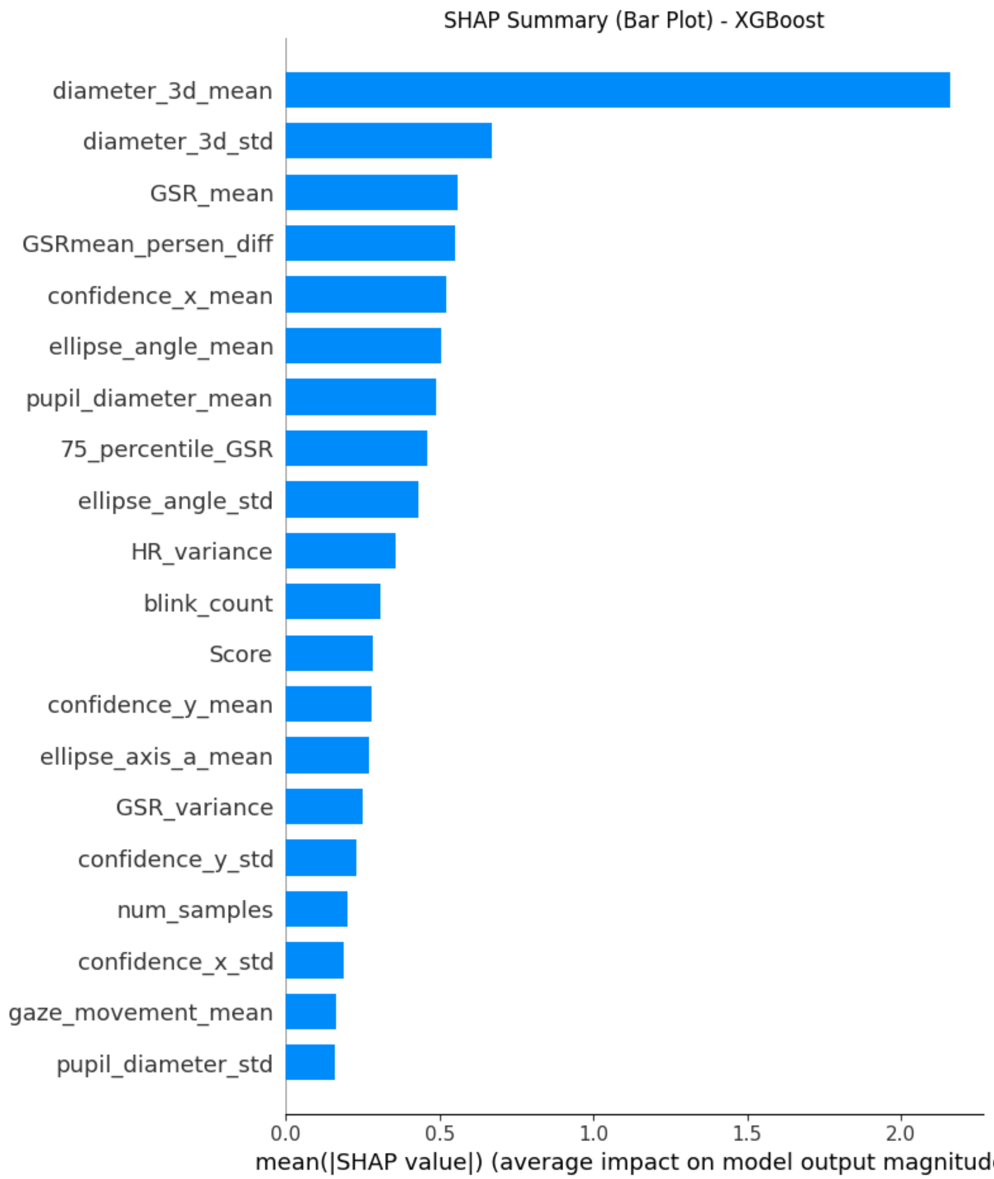

Figure 9: SHAP Summary (Bar Plot) for XGBoost

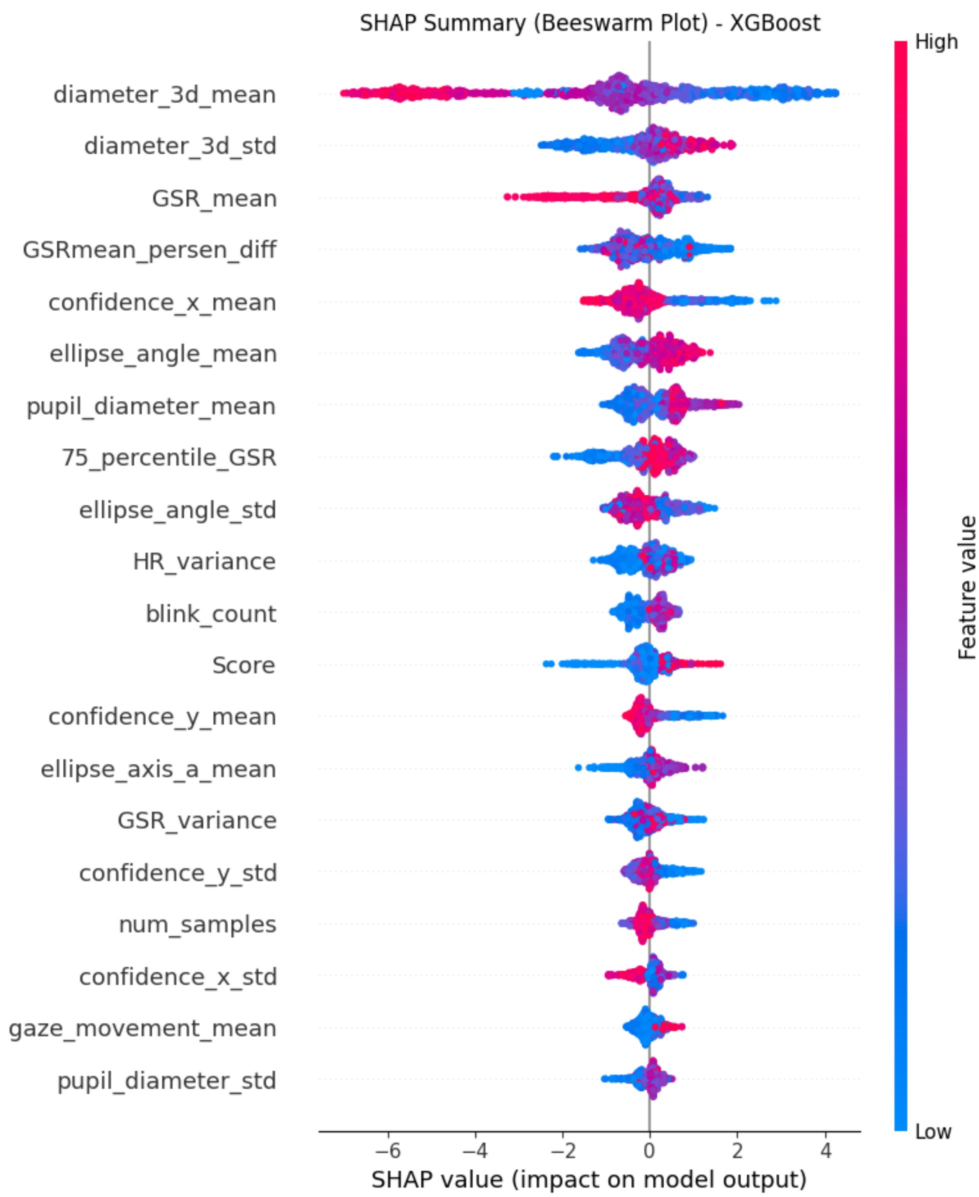

Figure 10: SHAP Beeswarm Plot for XGBoost

Top 5 SHAP-ranked Features (XGBoost):

- `diameter_3d_mean`
- `diameter_3d_std`
- `GSR_mean`
- `GSRmean_persen_diff`
- `confidence_x_mean`

Random Forest Results

To assess consistency across models, we also applied SHAP to a Random Forest classifier. Since SHAP for Random Forest outputs values for each class, we used the SHAP values corresponding to the positive class (i.e., low engagement).

Figure 11 displays the average absolute SHAP values across all features, while Figure 12 visualizes how individual feature values influence the model output.

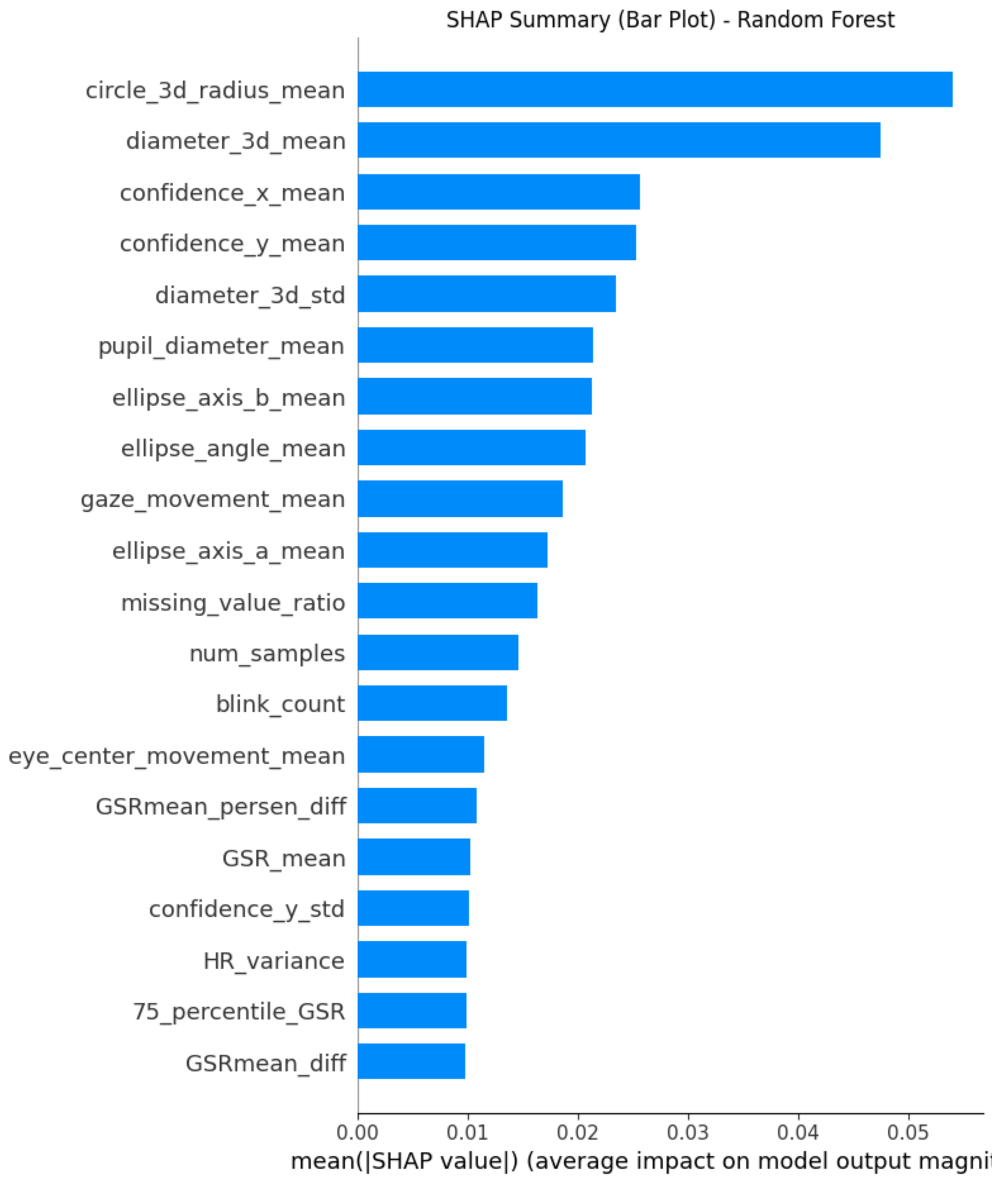

Figure 11: SHAP Summary (Bar Plot) for Random Forest

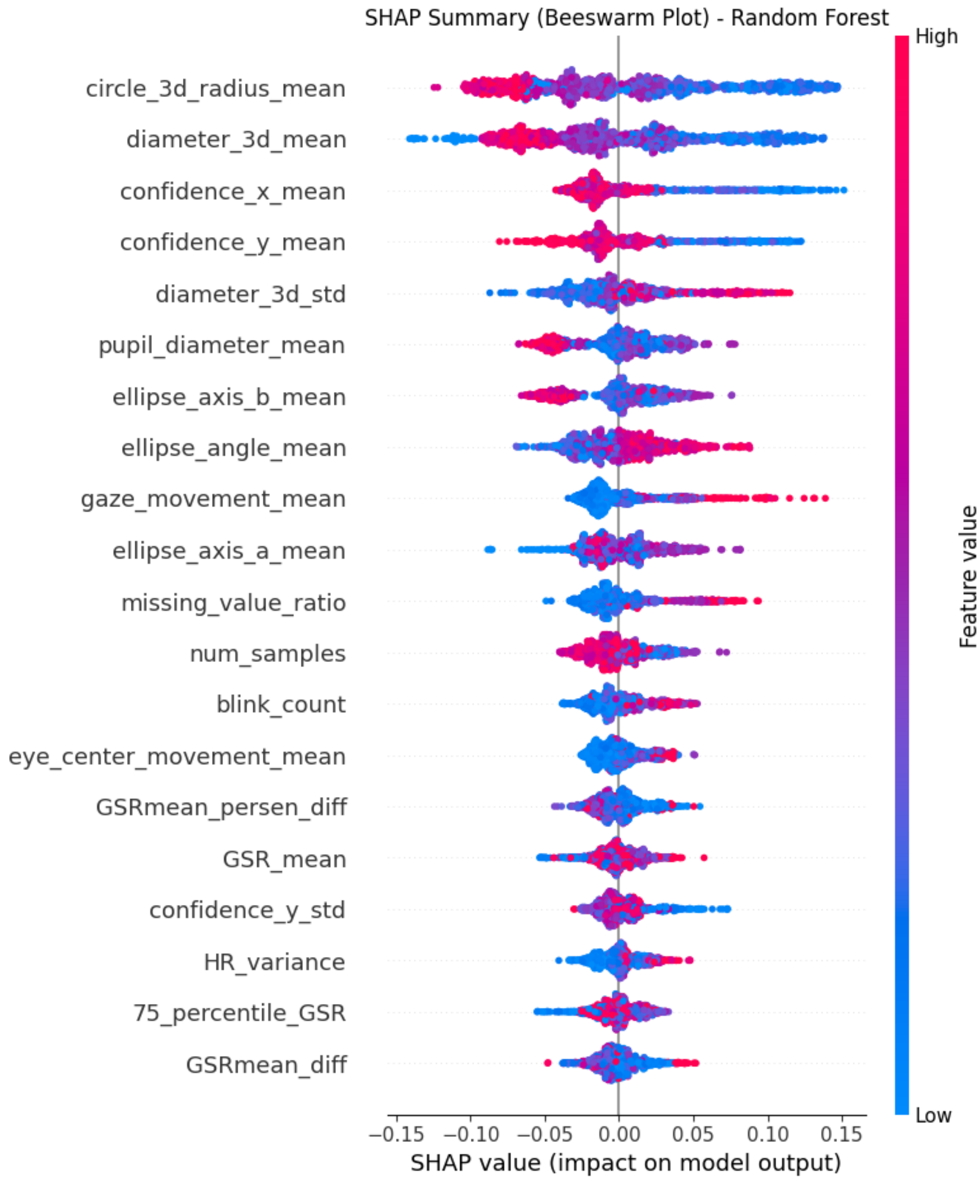

Figure 12: SHAP Beeswarm Plot for Random Forest

Top 5 SHAP-ranked Features (Random Forest):

- `circle_3d_radius_mean`
- `diameter_3d_mean`
- `confidence_x_mean`
- `confidence_y_mean`
- `diameter_3d_std`

Insights

These SHAP analyses across three different classifiers consistently highlight the predictive value of pupil diameter (3D), gaze confidence, and galvanic skin response (GSR) metrics. Their repeated appearance among the top features reinforces their importance in detecting low emotional engagement and complements the earlier correlation analysis.

4.4 Classical Feature Selection Methods

4.4.1 Mutual Information

To evaluate non-linear dependencies between features and the binary target label `probe`, we computed the Mutual Information (MI) scores using `mutual_info_classif` from `scikit-learn`. Figure 13 shows the bar plot of MI scores for all features.

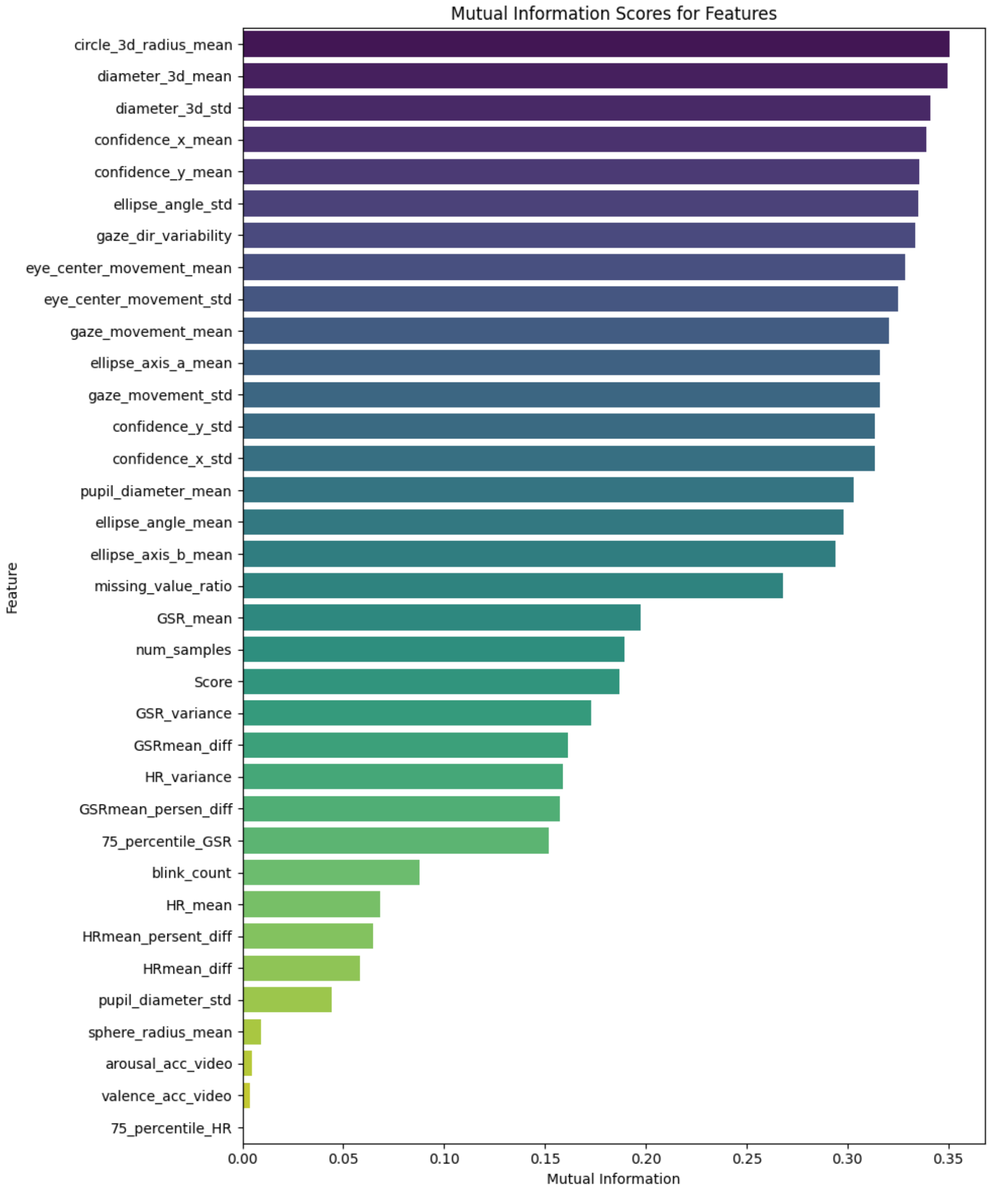

Figure 13: Mutual Information scores for all features with respect to the target label.

The top features based on MI scores are:

- `circle_3d_radius_mean` (0.351)
- `diameter_3d_mean` (0.350)
- `diameter_3d_std` (0.341)
- `confidence_x_mean` (0.339)
- `confidence_y_mean` (0.336)

4.4.2 ANOVA F-score Analysis

We also performed univariate feature selection using the ANOVA F-score method via `SelectKBest`. This method is useful to detect features that vary significantly across the binary classes. Figure 14 shows the F-scores for the top 20 features.

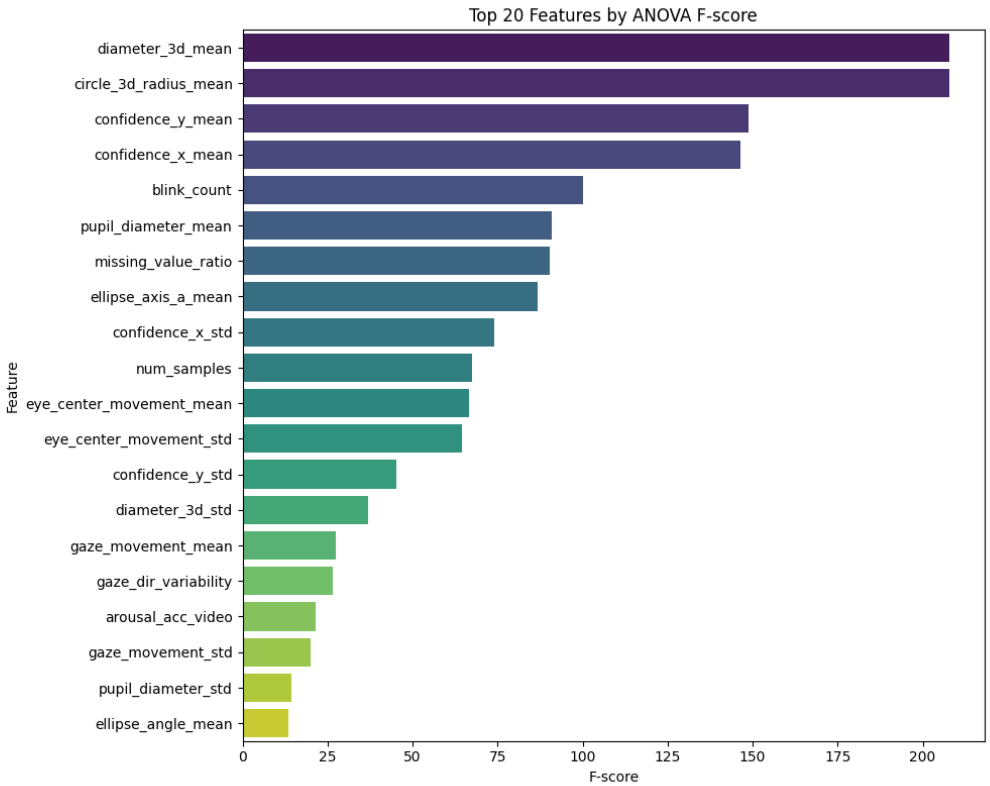

Figure 14: Top 20 features based on ANOVA F-score.

The top 5 features based on F-score and statistical significance are:

– `diameter_3d_mean` and `circle_3d_radius_mean` (F = 207.83, p ¡ 10^{-44})
– `confidence_y_mean` (F = 148.81, p ¡ 10^{-32})
– `confidence_x_mean` (F = 146.34, p ¡ 10^{-31})
– `blink_count` (F = 100.22, p ¡ 10^{-22})

These methods provide strong evidence that features derived from 3D pupil geometry and tracking confidence are the most discriminative in identifying low emotional engagement.

4.5 Multicollinearity and Feature Subset Selection

4.5.1 Variance Inflation Factor (VIF)

To assess multicollinearity among features, we computed the Variance Inflation Factor (VIF) for each standardized feature in the training set. VIF quantifies how much a feature is linearly predictable from the others.

The features `diameter_3d_mean` and `circle_3d_radius_mean` exhibited infinite VIF scores, indicating perfect multicollinearity. Additionally, several features had VIF values above 1000, such as:

- `confidence_x_mean`: VIF 2756
- `confidence_y_mean`: VIF 2784
- `GSR_mean`: VIF 893
- `75_percentile_GSR`: VIF 919

While multicollinearity may not affect tree-based models significantly, it can distort interpretability in linear models. Thus, VIF analysis informed our subsequent feature elimination strategy.

4.5.2 Recursive Feature Elimination with Cross-Validation (RFECV)

To identify the most informative and non-redundant features, we applied Recursive Feature Elimination with Cross-Validation (RFECV) using a logistic regression model with L2 regularization. Figure 15 illustrates the cross-validation accuracy as a function of the number of selected features.

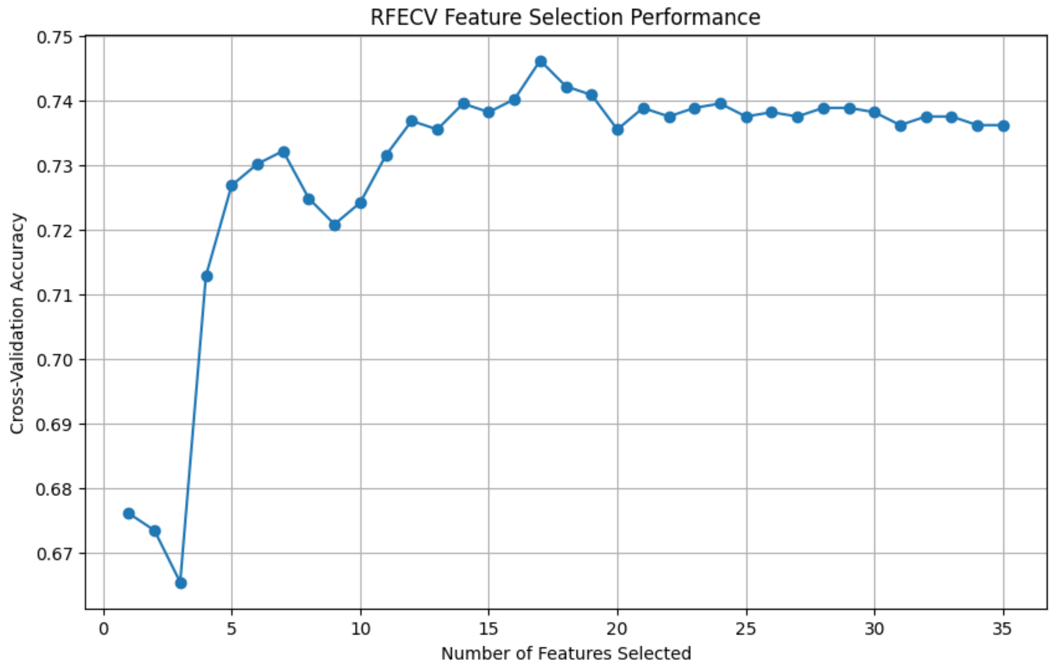

Figure 15: RFECV cross-validation accuracy vs number of features selected.

RFECV selected an optimal subset of 17 features out of the original 35. The top selected features include:

- `diameter_3d_mean`, `circle_3d_radius_mean`, `ellipse_axis_b_mean`
- `confidence_x_mean`, `confidence_y_mean`, `confidence_y_std`
- `blink_count`, `gaze_movement_mean`, `gaze_movement_std`
- `missing_value_ratio`, `arousal_acc_video`, `75_percentile_HR`

These results align well with earlier SHAP and ANOVA results, providing robust support for the importance of pupil geometry, eye movement stability, and tracking confidence as key indicators of engagement level.

4.6 Feature Importance and Projection Analysis

4.6.1 Coefficient Importance (Logistic Regression)

To further interpret the selected features from RFECV, we analyzed the absolute coefficients from a logistic regression model. Larger absolute coefficients indicate a stronger influence on the predicted probability of low engagement.

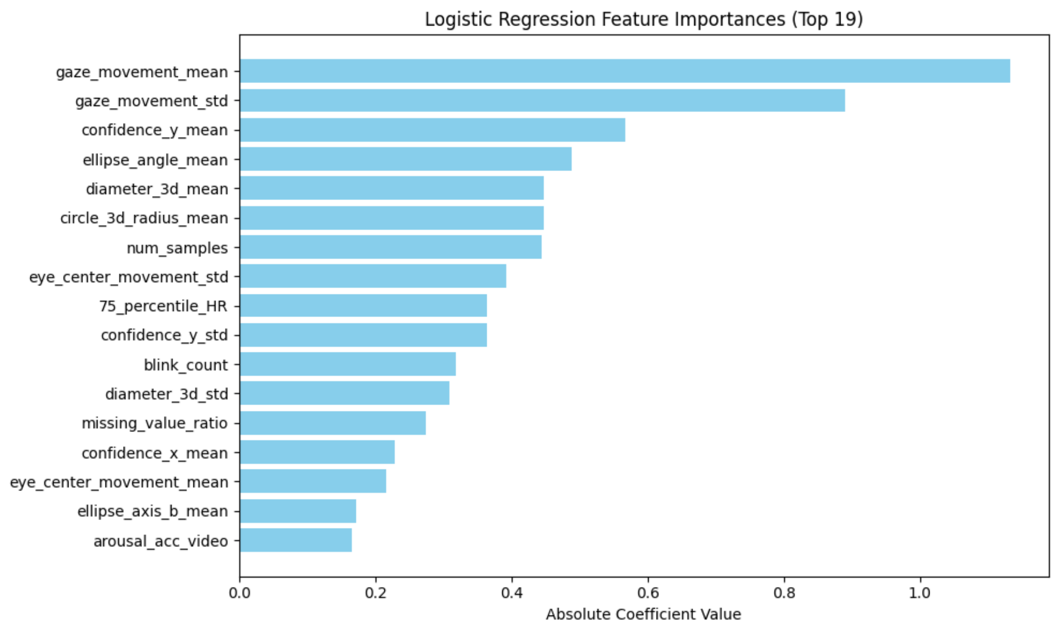

Figure 16: Absolute coefficients of selected features from Logistic Regression

The most influential features included:

- `gaze_movement_mean` (1.13)
- `gaze_movement_std` (0.89)
- `confidence_y_mean` (0.57)
- `ellipse_angle_mean` (0.49)
- `diameter_3d_mean` and `circle_3d_radius_mean` (0.45)

4.6.2 Random Forest Feature Importance

To validate the robustness of feature importance across model types, we trained a Random Forest classifier on the same 17 selected features. Feature importance was computed using mean impurity decrease.

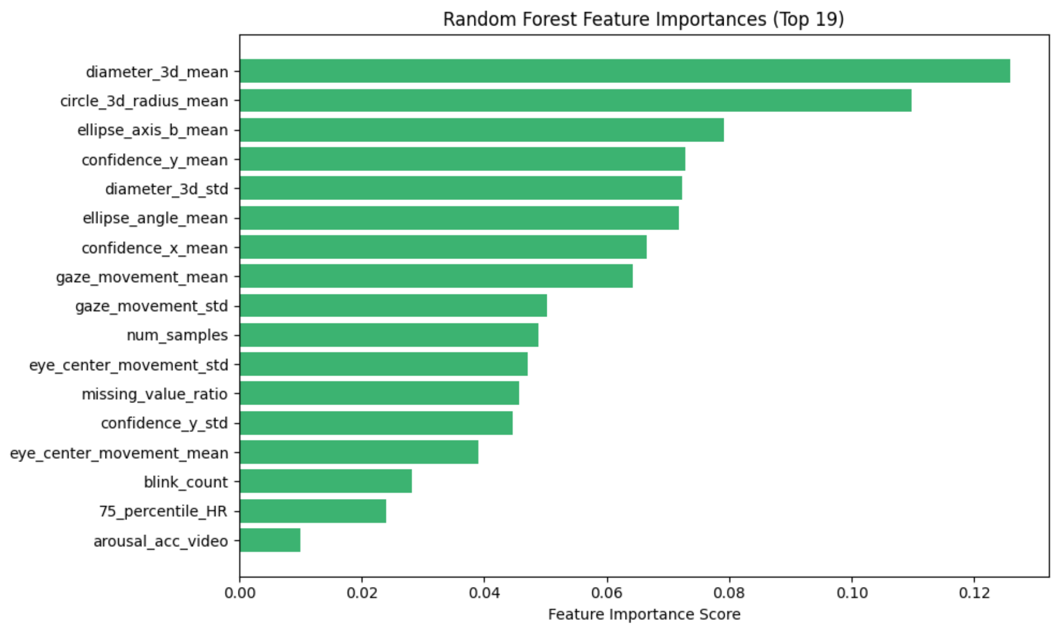

Figure 17: Random Forest Feature Importances

The top contributors were again `diameter_3d_mean` (0.126), `circle_3d_radius_mean` (0.110), and `gaze_movement_mean` (0.064), supporting consistency across both linear and tree-based models.

4.6.3 Principal Component Analysis (PCA)

To visualize separability in the selected features, we performed PCA on the standardized feature space. The first two principal components explained 46.25% of the total variance.

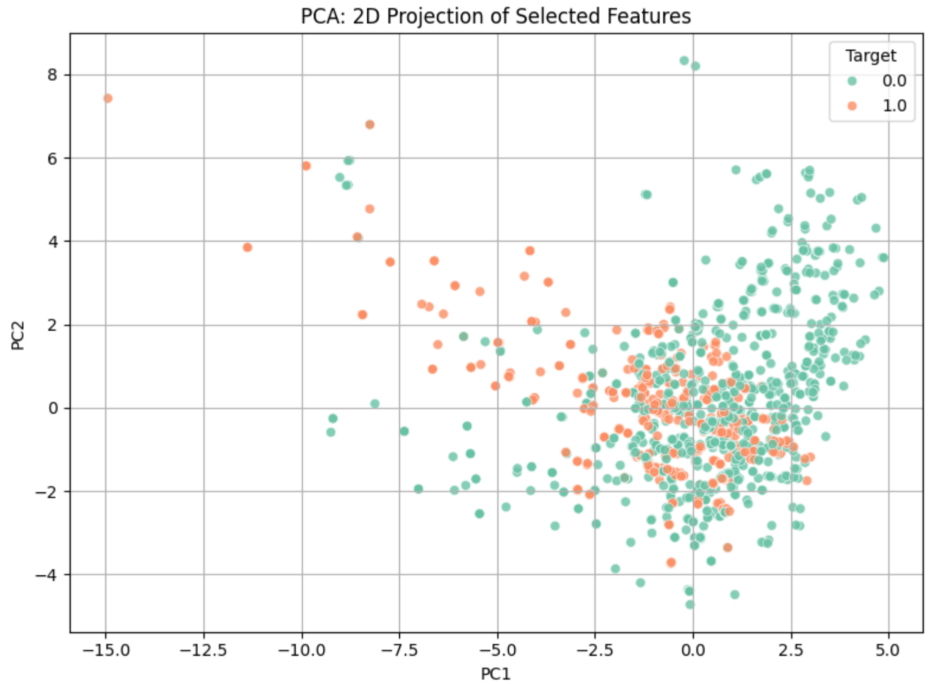

Figure 18: PCA 2D Projection using Selected Features

The projection shows partial clustering between high and low engagement labels, suggesting moderate discriminability using the selected physiological and eye-tracking features.

5 Preliminary Observations

In this section, we summarize key insights and trends obtained through exploratory modeling and feature importance analysis across multiple machine learning models. These observations are based on both global (model-wide) and local (instance-level) explanations of the multimodal features extracted from physiological signals and eye-tracking data.

5.1 Key Trends in Feature Importance

Across multiple models—namely Logistic Regression, Random Forest, XGBoost, and LightGBM—several features consistently emerged as important indicators of low emotional engagement.

- **Pupil diameter and eye geometry features**, such as `diameter_3d_mean`, `diameter_3d_std`, and `circle_3d_radius_mean`, were ranked among the top contributors by SHAP and Random Forest importance scores.

– **Eye-tracking confidence metrics**, including `confidence_x_mean` and `confidence_y_mean`, appeared frequently across top SHAP and model-based feature lists.

– **Gaze and eye movement features**—such as `gaze_movement_mean`, `gaze_movement_std`, and `eye_center_movement_std`—were particularly highlighted in Logistic Regression analysis.

– Physiological features like `GSR_mean` and `75_percentile_HR` showed moderate importance, indicating some relevance of physiological state in engagement levels.

5.2 Differences Across Users

Though model training was performed using aggregated data from four users, preliminary comparisons suggest certain feature distributions (e.g., blink rate, pupil diameter) show inter-user variability. This may reflect differences in baseline physiological and behavioral patterns or varied responses to the same stimuli. Such differences underscore the importance of accounting for user-specific baselines in personalized engagement detection models.

5.3 Dimensionality Reduction Insights

A 2D Principal Component Analysis (PCA) of the selected features revealed partial separation between engagement classes, with the first two principal components explaining approximately 46.25% of the total variance. This suggests that while the selected features carry meaningful information, engagement detection likely relies on more complex nonlinear relationships captured better by tree-based models.

5.4 Most Promising Features for Engagement Detection

Synthesizing results across SHAP, Logistic Regression coefficients, and Random Forest importance, the most promising features identified so far are:

– `diameter_3d_mean`

– `circle_3d_radius_mean`

– `confidence_x_mean`, `confidence_y_mean`

– `gaze_movement_mean`, `gaze_movement_std`

– `eye_center_movement_std`

These features are likely to play a central role in downstream modeling and real-time detection pipelines for low emotional engagement in multimodal settings. A consolidated summary of the top 10 features identified across all experiments is provided in the table below for reference.

Table 2: Top 10 Features Selected Across All Experiments

Rank	LightGBM (SHAP)	XGBoost (SHAP)	Random Forest
1	diameter_3d_mean	diameter_3d_mean	circle_3d_radius_mean
2	diameter_3d_std	diameter_3d_std	diameter_3d_mean
3	GSR_mean	GSR_mean	confidence_x_mean
4	GSRmean_persen_diff	GSRmean_persen_diff	confidence_y_mean
5	confidence_x_mean	confidence_x_mean	diameter_3d_std
6	HR_mean	confidence_y_mean	gaze_movement_mean
7	gaze_dir_variability	gaze_dir_variability	eye_center_movement_std
8	HR_variance	HRmean_diff	pupil_diameter_std
9	valence	pupil_diameter_mean	blink_count
10	ellipse_angle_mean	gaze_movement_std	ellipse_axis_a_mean

6 Conclusion and Next Steps

In this report, we presented the progress made across the first five phases of our project on detecting low emotional engagement using multimodal sensor data. We successfully conducted literature review and problem scoping, finalized scientifically validated video stimuli, built an integrated web platform for synchronized data collection, and gathered high-quality multimodal data from multiple users. Our preprocessing pipeline enabled alignment across physiological, eye-tracking, and self-reported annotations, and we carried out extensive exploratory and feature-level analysis to understand trends and variability.

The next immediate step is to initiate baseline model training and evaluation (Phase 6) using the engineered features and annotated labels. This will include benchmarking classifiers such as Random Forest, XGBoost, and LightGBM, and interpreting model outputs using SHAP and other explainability tools.

Following that, we plan to scale our data collection effort (Phase 7) to include 20 additional users. This will allow us to evaluate model generalizability, uncover deeper engagement patterns, and strengthen the robustness of our findings.

This work lays the foundation for building emotion-aware adaptive systems, and future phases will further refine our models and explore deployment in real-world settings.